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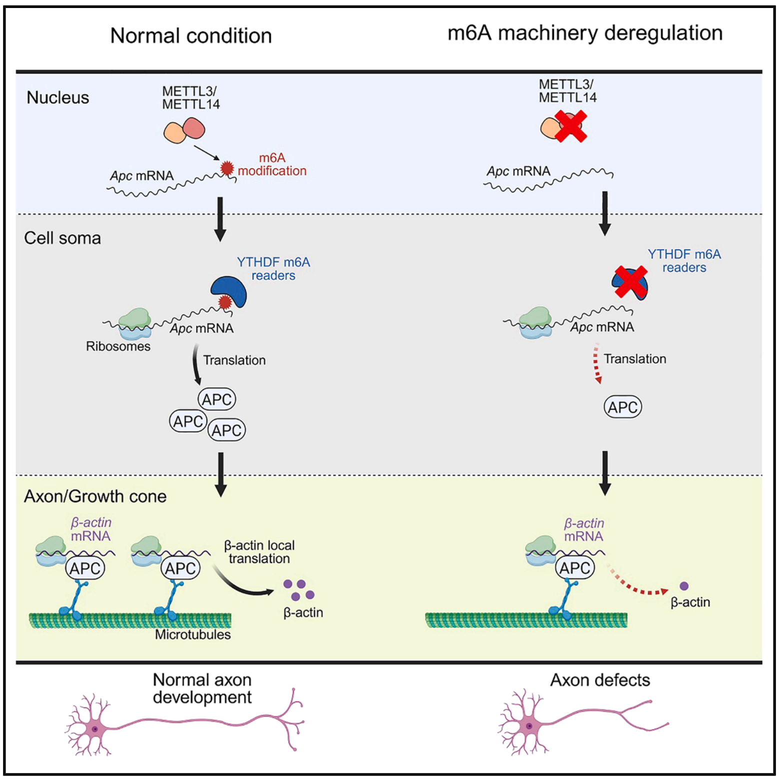
m^{6}A RNA methylation-mediated control of global APC expression is required for local translation of β-actin and axon development

Graphical abstract



Authors

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In brief

Broix et al. demonstrate that m^{6}A modifications on *Apc* mRNA facilitate its recognition by YTHDF readers, driving APC protein translation in the neuronal soma. The identified m^{6}A/YTHDFs/APC pathway regulates the axonal localization and local translation of β-actin mRNA essential for growth cone and axon development.

Highlights

- APC expression in the neuronal soma is m^{6}A dependent
- The m^{6}A machinery controls β-actin mRNA axonal transport and translation via APC expression
- m^{6}A/YTHDFs/APC pathway is required for axon development

Broix et al., 2025, Cell Reports *44*, 115727
June 24, 2025 © 2025 The Author(s). Published by Elsevier Inc.
https://doi.org/10.1016/j.celrep.2025.115727

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m^6A RNA methylation-mediated control of global APC expression is required for local translation of β-actin and axon development

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https://doi.org/10.1016/j.celrep.2025.115727

SUMMARY

The spatial regulation of mRNAs in neurons, including their localization and translation, is controlled by RNA-binding proteins and is critical for neuronal development. In this study, we present evidence that the multifunctional RNA-binding protein adenomatous polyposis coli (APC) is encoded by an mRNA modified with N6-methyladenosine (m^6A). This modification facilitates the translation of APC in neuronal somata via YTH domain-containing family (YTHDF) m^6A reader proteins. Disrupted APC expression, caused by reduced expression of the m^6A writer METTL14 or reader YTHDF1, or by overexpression of METTL14 mutants carrying human missense mutations linked to autism and schizophrenia, impairs the transport and local translation of APC-regulated target mRNA β-actin in axons and growth cones. Such disruptions consequently hinder axon development both *in vitro* and *in vivo*. These findings reveal a mechanism by which m^6A-regulated global expression of the RNA-binding protein APC governs axonal mRNA translation and development.

INTRODUCTION

Subcellular mRNA localization and local translation regulate gene expression spatially, especially in polarized cells like neurons. Numerous mRNAs are transported to distant axonal and dendritic compartments, where their local translation supports neuronal development and maturation.[1,2] At axonal growth cones, it facilitates axon guidance by enabling rapid cytoskeletal remodeling.[3] mRNA distribution in neurons relies on membrane-less ribonucleoprotein (RNP) granules, which interact with molecular motors for transport along microtubules.[4,5] These granules form through multivalent interactions between mRNAs with *cis*-acting motifs and *trans*-acting RNA-binding proteins (RBPs).[6]

While known as a tumor suppressor, adenomatous polyposis coli (APC) also functions as an RBP in migrating fibroblasts, *Drosophila* oocytes, and neurons.[7–9] APC binds to the 3′UTR of *Tubb2b* mRNA and other cytoskeletal regulators, promoting their axonal localization and translation to regulate microtubule dynamics in growth cones.[8] It also links β-actin and *Tubb2b* mRNAs to kinesin-2 for microtubule transport.[10] Beyond its role as an RBP, APC exhibits direct interaction with the microtubule lattice and can bind to the growing microtubule ends through its interaction with end-binding proteins, which are a core family of microtubule plus-end tracking proteins that have profound effects on the shape of the microtubule network.[11,12] Moreover, APC can bind actin and nucleate unbranched actin filaments either independently or in collaboration with formins.[13–15] Reflecting its cytoskeletal functions, APC is crucial for cell motility, polarity, and neurite outgrowth and is implicated in neurological disorders like autism and schizophrenia.[16–25]

RNA modifications have emerged as key post-transcriptional regulators in the central nervous system.[26,27] N6-methyladenosine (m^6A) is the best characterized and the most abundant internal mRNA modification in mammals, which decorates up to half of the entire poly(A) transcriptome in the brain tissue.[28] m^6A plays essential roles in neurogenesis and differentiation,[29–32] axon guidance and regeneration,[33–39] synaptic function,[40,41] and memory.[42–46] Advances in m^6A research stem from identifying its "writers" (METTL3, METTL14, and WTAP), "erasers" (FTO and ALKBH5), and "readers" (YTH proteins, hnRNPs, IGF2BPs, and eIF3). YTH N6-methyladenosine RNA binding proteins (YTHDF) readers regulate gene expression by influencing mRNA export, stability, translation, and phase separation.[32,47–54] We previously found that hypermethylated synapse-related transcripts are enriched at synapses, suggesting a role for m^6A in mRNA localization and local translation.[40] This aligns with findings that learning-related m^6A-modified RNAs localize to synapses for memory consolidation.[55] Other studies,

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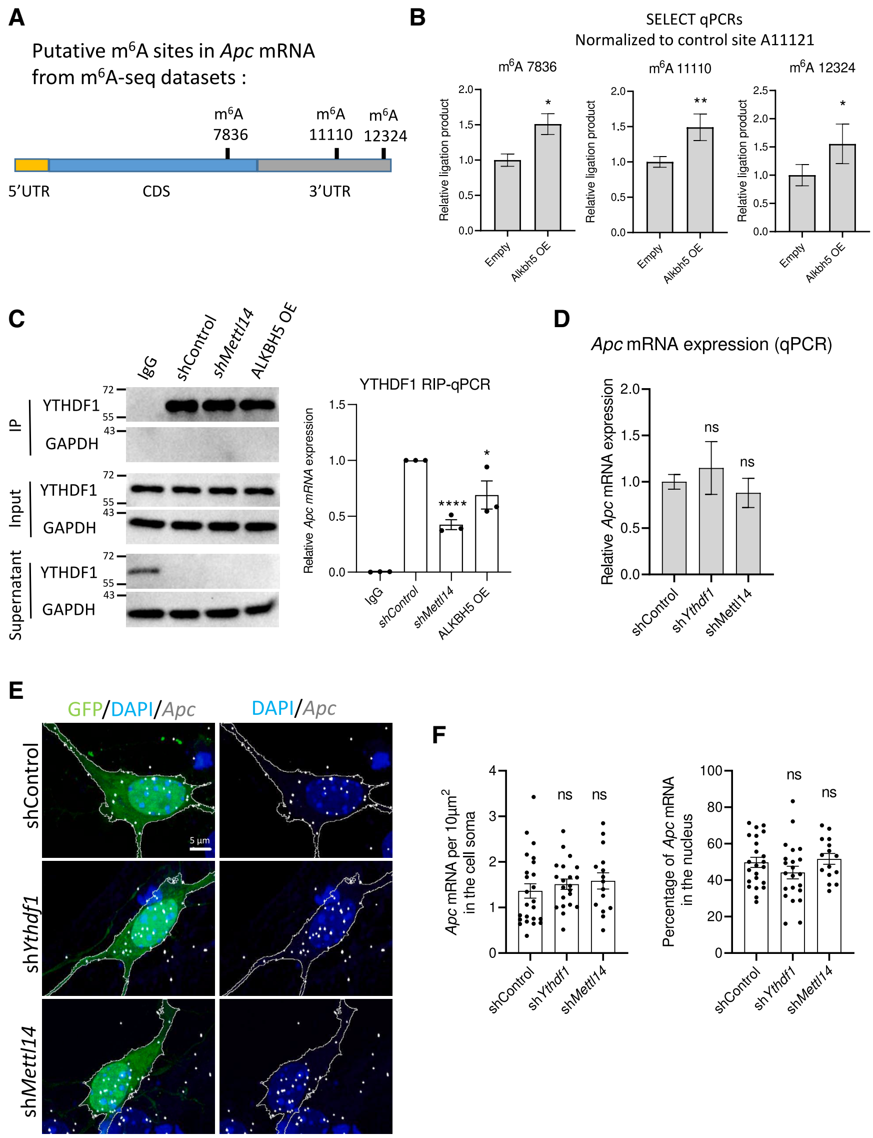


Figure 1. *Apc* mRNA is m^{6}A modified and bound by YTHDF1, but this does not impact its mRNA expression or nuclear export

(A) Schematic representation of *Apc* mRNA (5′UTR in yellow, CDS in blue, and 3′UTR in gray) with three putative m^{6}A-modified nucleotides to be validated by SELECT.

(B) Measurement of m^{6}A level by SELECT on A7836, A11110, and A12324 of *Apc* mRNA in Neuro2a cells transfected with an empty vector or an ALKBH5-overexpressing (OE) vector. Ligation products obtained for A7836, A11110, and A12324 were normalized to *Apc* mRNA abundance by performing SELECT with the control site A11121 (A12324: empty, $n = 4$; ALKBH5 OE, $n = 4$; A11110: empty, $n = 4$; ALKBH5 OE, $n = 4$; A7836: empty, $n = 3$; ALKBH5 OE, $n = 3$).

(C) RIP-qPCR analysis of *Apc* mRNA bound by YTHDF1; the left panel shows the efficiency of YTHDF1 immunoprecipitation (IP) in immunoglobulin G (IgG), shControl, sh*Mettl14*, and ALKBH5 OE. *Apc* mRNA expression is relative to shControl, and *Apc* mRNA abundance was normalized to *Gapdh* in input RNA, $n = 3$.

(D) qPCR quantification of *Apc* mRNA normalized to *Gapdh* obtained from Neuro2a cells transfected with shControl, sh*Ythdf1*, or sh*Mettl14* (shControl, $n = 10$; sh*Ythdf1*, $n = 10$; sh*Mettl14*, $n = 9$).

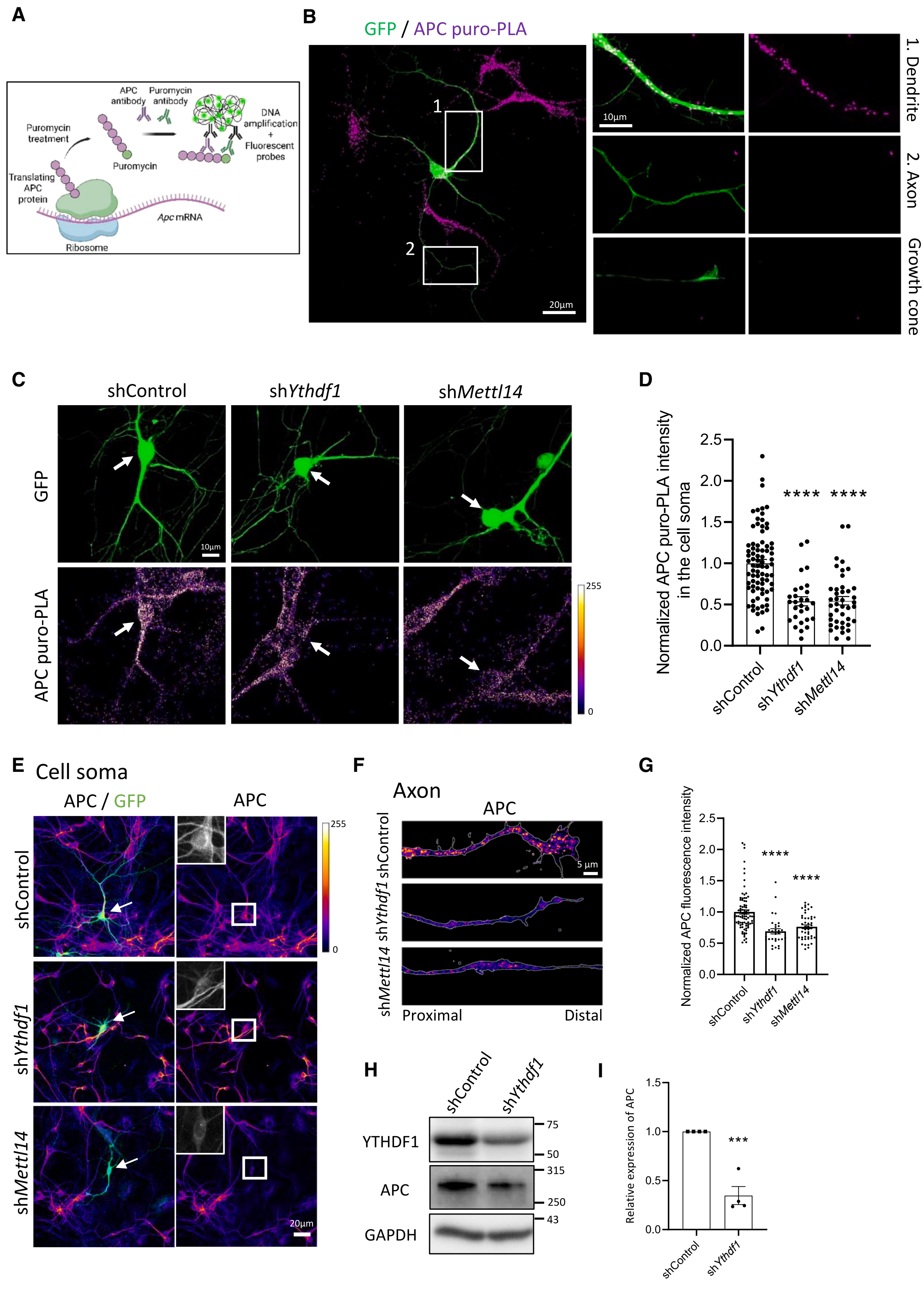
(E) Representative images of *Apc* ViewRNA ISH (white) in DIV6 neurons transfected with shControl, sh*Ythdf1*, or sh*Mettl14* together with a GFP-expressing plasmid (green) and counterstained with DAPI (blue). Scale bar, 5 μm.

(F) Quantification of the number of *Apc* mRNA puncta in the cell soma (left, number of neurons: shControl, $n = 24$; sh*Ythdf1*, $n = 21$; sh*Mettl14*, $n = 15$) and the percentage of *Apc* mRNA in the nucleus (right, number of neurons: shControl, $n = 23$; sh*Ythdf1*, $n = 22$; sh*Mettl14*, $n = 15$).

The graphs represent the mean ± SD (B and D) and the mean ± SEM in (C and F). $^*p < 0.05$, $^{**}p < 0.01$, and $^{****}p < 0.0001$ or ns (not significant) comparing conditions to each other using one-way ANOVA tests (C, D, and F) and unpaired t test (B). See also Figure S1.

however, have found conflicting results regarding the involvement of m^{6}A signaling in controlling mRNA localization in neurons. Flamand and Meyer reported that m^{6}A methylation in the 3′UTR of hundreds of mRNAs facilitates their distal transport to neurites,[56] whereas Loedige et al. described that m^{6}A-modified transcripts are predominantly restricted to neuronal cell soma due to reduced stability.[57] Further research is needed to clarify m^{6}A's role in mRNA localization. Our previous m^{6}A-sequencing analysis identified *Apc* mRNA among the most m^{6}A-methylated transcripts in neurons, but its functional significance remains unknown.[40] A cell biological functional link between the m^{6}A machinery and the expression control of RBPs, a process crucial for axonal development that engages local protein synthesis, could provide insights into the functional mechanism of m^{6}A.

Here, we provide evidence of m^{6}A-mediated regulation of APC translation that modulates cytoskeleton dynamics at the axonal growth cones. We report that m^{6}A modifications are installed on *Apc* mRNA and that the subsequent recognition and binding by YTHDF readers regulate APC translation within neuronal somata. We further show that interfering m^{6}A pathway disrupts the transport and local translation of β-actin mRNA in the axon and growth cone without affecting its translation in soma, which can be rescued by exogenously expressed APC protein. We validate the functional requirement of YTHDF1 for commissural axon development in the corpus callosum during cortical development. Furthermore, we confirm the biological relevance of the m^{6}A/YTHDFs/APC pathway by showing that mutations in METTL14, which are associated with autism and schizophrenia, disrupt *Apc* mRNA binding by YTHDF1, leading to decreased expression of APC and axon growth defects. Together, our data reveal that the m^{6}A-mediated control of APC translation in the cell soma is responsible for β-actin axonal mRNA trafficking



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and local translation that are required for the progression of axon development.

RESULTS

m^{6}A modification is installed on *Apc* mRNA but does not impact *Apc* mRNA stability or nuclear export in neurons

Previous studies, including our own, identified *Apc* mRNA as highly m^{6}A modified,[40,45,52,58,59] but the modification sites remained unvalidated. We employed an antibody-independent m^{6}A detection method named SELECT (single-base elongation- and ligation-based PCR amplification) to confirm selective m^{6}A marks on *Apc* mRNA[60] (Figure S1A). Three adenosines in mouse *Apc* mRNA (GenBank: NM_007462.3) were chosen for validation, with one in the coding sequence (A7836) and two in the 3′UTR (A11110 and A12324), as predicted by high-throughput analysis (Figure 1A). SELECT amplification for the three sites relative to a control adenosine site revealed a significant increase in ligation products for all three sites in the cells transfected with an ALKBH5-overexpression plasmid compared to the control, confirming these three adenosines as m^{6}A-methylated sites in *Apc* mRNA (Figure 1B). Notably, ALKBH5 overexpression did not alter *Apc* mRNA levels (Figure S1B). To investigate m^{6}A's role in *Apc* mRNA regulation, we analyzed the effects of YTHDF1 and METTL14 knockdowns in cultured neurons. The consensus is that YTHDF1 regulates translation, YTHDF2 predominantly facilitates degradation, and YTHDF3 regulates both transcript stability and translation through interactions with either YTHDF1 or YTHDF2.[61] As we did not observe any alteration in *Apc* mRNA expression following ALKBH5 overexpression, we selected YTHDF1 among the three YTHDF proteins for further investigation. METTL14 was selected over METTL3 due to the high level of toxicity observed in neuronal cells following METTL3 knockdown. Short hairpin RNAs (shRNAs) against *Ythdf1* and *Mettl14* effectively reduced their mRNA and protein levels, validated via qPCR and immunostaining in days *in vitro* (DIV) 6 neurons transfected at DIV2 (Figures S1C–S1F). To determine whether YTHDF1 binds to *Apc* mRNA in an m^{6}A-dependent manner, we conducted RNA immunoprecipitation (RIP) analysis in Neuro2a cells. RIP-qPCR results revealed that YTHDF1 binds to *Apc* mRNA (Figure 1C). Furthermore, knockdown of METTL14 or overexpression of ALKBH5 significantly reduced the amount of *Apc* mRNA bound to YTHDF1, indicating that m^{6}A modification is required for this interaction (Figure 1C). Following this validation, we performed qPCR analysis to determine *Apc* mRNA expression in Neuro2a cells transfected with either sh*Ythdf1* or sh*Mettl14*, and we observed no difference in the expression of *Apc* mRNA in comparison to the control (Figure 1D). We utilized ViewRNA *in situ* hybridization (ISH) method to detect *Apc* mRNA in cultured DIV6 neurons, and we observed that YTHDF1 or METTL14 downregulations had no impact on the density of *Apc* mRNAs in the cell soma and on the nucleus/cytosol ratio (Figures 1E and 1F). Overall, these data indicate that *Apc* mRNA is m^{6}A modified and bound by m^{6}A reader YTHDF1, but m^{6}A is not essential for the expression, stability, and nuclear export of *Apc* mRNA in neurons.

APC protein synthesis in the neuronal somata is controlled by the m^{6}A machinery

Since YTHDF1 and METTL14 deregulation did not affect *Apc* mRNA expression or export, we investigated whether m^{6}A modification regulates APC protein translation. Using a proximity ligation assay (PLA) in puromycin-treated neurons, where anti-puromycin and anti-APC antibodies were utilized, we visualized *de novo* APC protein synthesis events, as indicated by puro-PLA puncta (Figure 2A).[62] In control neurons, APC translation occurred in the soma and dendrites but not in axons/growth cones, suggesting that APC function in this region primarily relies on the control of its protein transport, rather than local translation (Figure 2B). Axons and dendrites were identified using Tau and MAP2 staining, respectively, and we verified the specificity of the APC puro-PLA staining (Figures S2A–S2C). We next examined the impact of YTHDF1 and METTL14 knockdowns on APC protein synthesis in the cell soma. YTHDF1 or METTL14 knockdown significantly reduced APC protein synthesis in the soma, indicating that m^{6}A modification of *Apc* mRNA and its recognition by YTHDF1 are essential for proper APC translation (Figures 2C and 2D). Immunostaining and western blot analyses further confirmed that knocking down YTHDF1 or METTL14 at DIV2 reduced APC fluorescence in soma and neurites by DIV6 (Figures 2E–2G) and decreased APC protein levels in Neuro2a cells (Figures 2H and 2I). We next sought to understand whether the m^{6}A-mediated control of APC translation is

Figure 2. The m^{6}A machinery regulates APC protein synthesis in the neuronal somata

(A) Illustration of the puro-PLA assay used to detect the newly synthesized APC protein.

(B) Representative image of APC puro-PLA labeling (purple) in a DIV6 neuron expressing GFP (green). Scale bar, 20 μm. (1) and (2) provide higher magnifications of the boxed areas from the left panel located in a primary dendrite and in the axon, respectively. The right panel shows a higher magnification of the corresponding axonal growth cone. Scale bar, 10 μm.

(C) Representative images showing newly synthesized APC puro-PLA signal (Fire Look Up Table [LUT]) in DIV6 neurons transfected with shControl, sh*Ythdf1*, or sh*Mettl14* along with GFP (green), treated with puromycin for 5 min. Transfected neurons are indicated by arrows. Scale bar, 10 μm.

(D) Quantification of APC puro-PLA fluorescence signal intensity in the cell soma (number of cells: shControl, n = 80; sh*Ythdf1*, n = 28; sh*Mettl14*, n = 42). The results are normalized to the average of the shControl.

(E and F) Representative images of cell soma (E) and axons (F) from neurons transfected with shControl, sh*Ythdf1*, or sh*Mettl14* along with GFP, labeled with an anti-APC antibody (Fire LUT). The insets show a higher magnification of APC staining in the cell soma of transfected neurons. Scale bar, 20 μm (E) and 5 μm (F).

(G) Quantification of APC fluorescence intensity in the cell soma and neurites of neurons transfected with shControl (number of neurons: n = 84), sh*Ythdf1* (n = 31), or sh*Mettl14* (n = 46). The results are normalized to the average of the shControl.

(H and I) Representative immunoblots (H) and quantitative analysis (I). GAPDH was used as a loading control for normalization (shControl, n = 4; sh*Ythdf1*, n = 4). The graphs represent the mean ± SEM in (D), (G), and (I). ***$p < 0.001$, and ****$p < 0.0001$ or ns (not significant) comparing conditions to each other using one-way ANOVA tests (D and G) and unpaired t test (I). See also Figures S2 and S3.

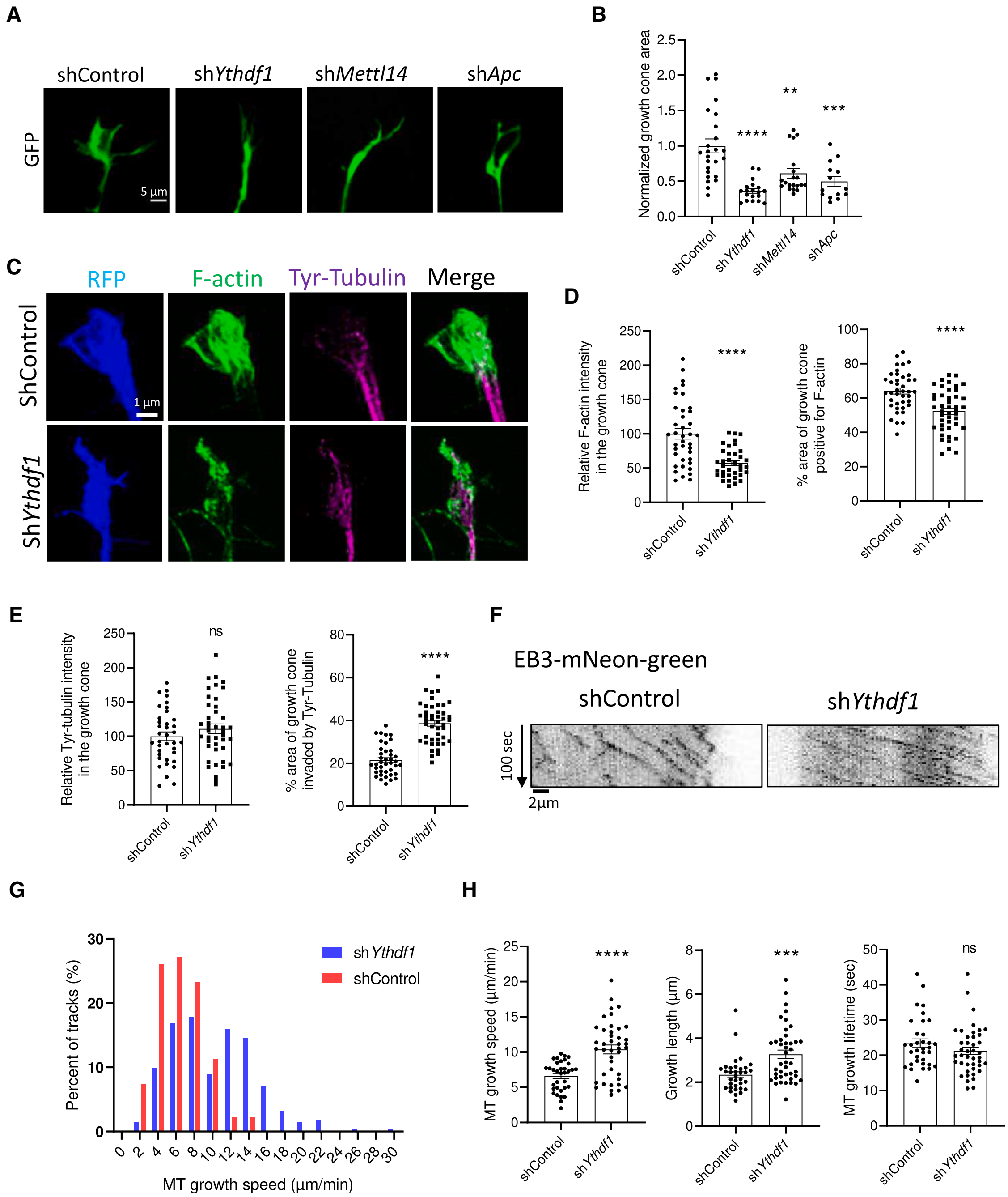

Figure 3. YTHDF1 downregulation induces loss of F-actin and defects in microtubule dynamics in the axonal growth cone

(A) Representative images showing the growth cone of DIV6 neurons transfected with GFP and the shControl, sh*Ythdf1*, sh*Mettl14*, or sh*Apc* at DIV2. Scale bar, 5 µm

(B) Quantification of the growth cone area (number of growth cones: shControl, *n* = 25; sh*Ythdf1*, *n* = 19; sh*Mettl14*; *n* = 20; sh*Apc*, *n* = 14). The results are normalized to the average of the shControl.

(C) Representative images of growth cones from DIV6 neurons expressing shControl or sh*Ythdf1* along with RFP (blue) and labeled with phalloidin (F-actin, green) and an anti-tyrosinated-tubulin antibody (Tyr-Tubulin, purple). Scale bar, 1 µm.

(D) Quantifications of the F-actin intensity in the growth cone (number of growth cone: shControl, *n* = 39; sh*Ythdf1*, *n* = 36) and the percentage of the growth cone area positive for F-actin (number of growth cone: shControl, *n* = 40; sh*Ythdf1*, *n* = 45).

(E) Quantification of the Tyr-tubulin fluorescence intensity in the growth cone (number of growth cones: shControl, *n* = 37; sh*Ythdf1*, *n* = 42) and the percentage of the growth cone area invaded by Tyr-tubulin (number of growth cones: shControl, *n* = 40; sh*Ythdf1*, *n* = 45).

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specific to YTHDF1 protein in the YTHDF family. RIP-qPCR analysis revealed that YTHDF2 and YTHDF3 also bind to *Apc* mRNA in an m^{6}A-dependent manner (Figures S3A and S3B). We knocked down YTHDF2 and YTHDF3 independently in Neuro2a cells and found reduced APC protein expression similar to what we observed after YTHDF1 knockdown (Figures S3C and S3D). APC puro-PLA assays showed a significant, though less pronounced, reduction in newly synthesized APC protein upon YTHDF2 or YTHDF3 knockdown (Figures S3E and S3F), while *Apc* mRNA levels remained unchanged (Figure S3G). These findings suggest that all three YTHDF readers are involved in regulating APC protein translation in the neuronal cell soma. Immunoprecipitation in Neuro2a cells showed that YTHDF1 interacts with YTHDF2 and YTHDF3, suggesting coordinated regulation of *Apc* translation (Figure S3H). Further studies are needed to clarify the roles, interactions, and potential functional redundancy among YTHDF1 and other YTH family proteins in regulating translation. Altogether, these findings highlight the crucial role of the m^{6}A machinery in facilitating proper APC protein translation and maintaining its expression in neurons.

Downregulation of YTHDF1 disrupts microtubule and actin dynamics in the axonal growth cone

APC, which localizes in axonal growth cones, is crucial for regulating growth cone dynamics and axon elongation.[8,19,63–65] To validate the significance of APC in growth cone dynamics, we cultured dissociated mouse hippocampal neurons and transfected them with a validated shRNA against *Apc* (Figure S4A) or shControl at DIV2, along with a GFP-expressing plasmid, and analyzed growth cone area and shape at DIV6. Downregulation of APC resulted in a decreased growth cone area, with a transition from a fan-shaped structure to a collapsed tip (Figures 3A and 3B). Similar defects were observed with YTHDF1 or METTL14 knockdowns, suggesting a shared pathway regulating growth cone morphology (Figures 3A and 3B). To verify the specificity of axon defects and rule out effects on other cytoskeletal regulators, we examined collapsin response mediator protein 2 (CRMP2) and doublecortin (DCX) levels in neuronal subcompartments after YTHDF1 downregulation and found no significant changes (Figures S4B and S4C). APC's role in regulating the actin and microtubule cytoskeleton in the growth cone, demonstrated by its downregulation leading to decreased F-actin signal and increased microtubule growth rate in U2OS cells,[65,66] prompted us to investigate whether similar defects would occur following YTHDF1 deregulation. Staining for F-actin and tyrosinated microtubules revealed that YTHDF1 downregulation decreased the signal and area of F-actin signal in the growth cone, while increasing tyrosinated microtubules invasion (Figures 3C–3E). Live imaging of EB3-NeonGreen comets, which track microtubule growing ends, showed that YTHDF1 downregulation increased microtubule growth speed and length, with no effect on the duration of growth events (Figures 3F–3H). Collectively, these data demonstrate a novel role for the m^{6}A machinery in governing the actin and microtubule cytoskeleton in the growth cone, which is likely to be mediated through the control of APC protein expression.

The m^{6}A/YTHDF1/APC pathway regulates β-actin mRNA transport and local translation in the axon

To investigate how the m^{6}A machinery and APC regulate growth cone dynamics, we examined β-actin mRNA transport and local translation for the following reasons: APC has been shown to modulate β-actin mRNA transport by acting as a scaffold between molecular motors and β-actin mRNA,[10,67] and our results indicate that YTHDF1 knockdown reduces the F-actin signal in the growth cone (Figures 3C and 3D). Using the ViewRNA method to detect β-actin mRNA in neurons transfected with shRNA against *Ythdf1*, *Mettl14*, or the control, we found that YTHDF1 knockdown did not affect β-actin mRNA density in the soma or its nuclear export. However, METTL14 downregulation increased total β-actin mRNA levels without altering nuclear export, suggesting an m^{6}A-dependent destabilization mechanism (Figures S5A–S5C). Due to technical constraints in quantifying β-actin mRNA in the axon (e.g., dim GFP fluorescence after ViewRNA ISH treatments), we performed live imaging of fluorescently labeled β-actin mRNA using the PP7-based labeling system in axons (Figure 4A). YTHDF1 downregulation did not affect β-actin mRNA axonal transport dynamics, including average speed, cumulative distance, instantaneous speed, or the percentage of moving β-actin mRNA granules (Figures 4A and 4B). However, we observed that YTHDF1 knockdown led to a decrease of the β-actin mRNA granule density in the axon, indicating that less β-actin mRNA is transported to the axon compared to the control (Figures 4A and 4B). Importantly, APC knockdown similarly reduced β-actin mRNA granule density in the axon without altering its transport dynamics (Figures S5D and S5E). To investigate why the transport dynamics of the remaining β-actin mRNA granules were unaffected by YTHDF1 knockdown, we performed APC immunostainings on neurons transfected with the fluorescently labeled β-actin mRNA construct. We found that APC/β-actin mRNA colocalization density was reduced in axons of neurons transfected with sh*Ythdf1* compared to shControl (Figures 4C and 4D). However, the percentage of β-actin mRNA granules colocalized with APC remained unchanged (shControl: 66.78%; sh*Ythdf1*: 62.55%) (Figures 4C and 4D), suggesting that the remaining β-actin mRNA granules observed after YTHDF1 knockdown were still bound by APC, maintaining their transport dynamics. To determine if reduced β-actin mRNA in the axon affects local translation upon YTHDF1 downregulation, we analyzed *de novo* β-actin

(F) Representative kymographs of live imaging of EB3-NeonGreen comets in the growth cone of neurons transfected with shControl or sh*Ythdf1*. The *x* axis corresponds to the position of the particles (μm), and the *y* axis to the time (seconds, from top to bottom). Scale bar, 2 μm.

(G) Graph representing the distribution of microtubule growth speed in shControl and sh*Ythdf1* conditions.

(H) Quantification of the microtubule growth speed, microtubule growth length, and microtubule growth lifetime (number of comets: shControl, $n = 34$; sh*Ythdf1*, $n = 42$).

The graphs represent the mean ± SEM in (B, D, E, and H). $^{***}p < 0.001$, and $^{****}p < 0.0001$ or ns (not significant) comparing conditions to each other using one-way ANOVA tests (B) and unpaired t test (D, E, and H). See also Figure S4.

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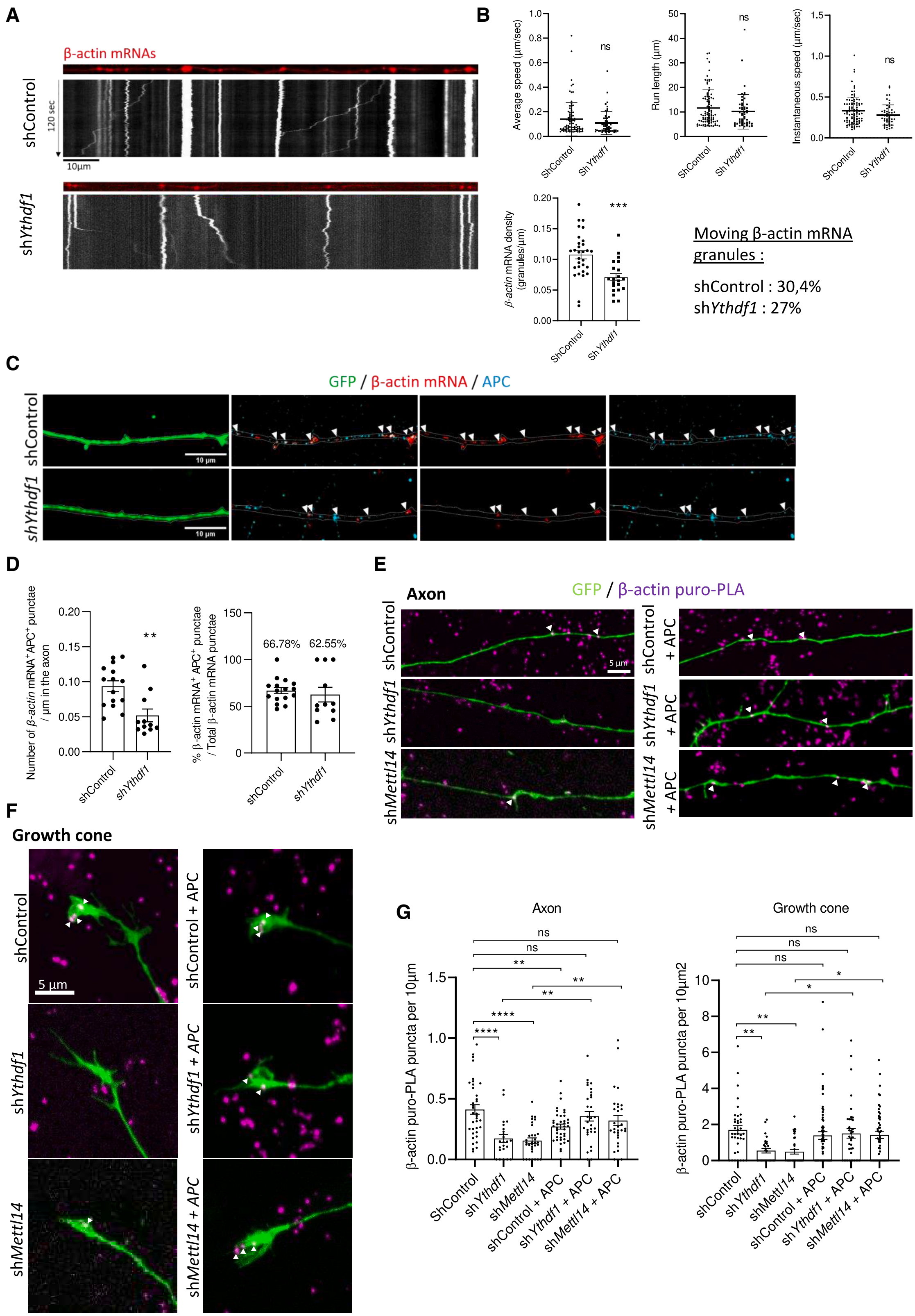

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synthesis using the puro-PLA method. Immunofluorescence and quantitative analysis showed that YTHDF1 or METTL14 knockdown significantly decreased newly synthesized β-actin puncta in the axon shaft and growth cone, while soma synthesis remained unchanged (Figures 4E–4G and S5F–S5I). To confirm that impaired axonal β-actin translation is due to m^6A-dependent APC regulation, we performed rescue experiments by assessing β-actin puro-PLA signal in axons of neurons transfected with a hemagglutinin (HA)-tagged APC plasmid along with shControl, sh*Ythdf1*, or sh*Mettl14*. We first validated that HA-APC expression under a strong promoter compensates for APC loss induced by YTHDF1 deregulation (Figure S5J). APC overexpression in sh*Ythdf1*- and sh*Mett14*-transfected neurons restored β-actin local translation in the axon and growth cone to normal levels (Figures 4E–4G). Another APC-bound axonal mRNA crucial for cytoskeletal dynamics is *Tubb2b*.[8] Using the puro-PLA method, we found that YTHDF1 knockdown did not affect TUBB2B local translation in the axon or growth cone (Figures S6A–S6D) nor alter *Tubb2b* mRNA transport dynamics or granule density (Figures S6E and S6F). An *in vitro* study showed that APC and fragile X messenger ribonucleoprotein 1 (FMRP) bind *Tubb2b* mRNA with similar affinity, while FMRP binds *β-actin* mRNA less efficiently.[67] Notably, YTHDF1 knockdown did not change axonal or dendritic FMRP levels (Figure S7), suggesting that FMRP might compensate for *Tubb2b* transport but not *β-actin*. Altogether, these data suggest that the m^6A/YTHDF1/APC pathway is required for the transport of β-actin mRNA into the axon and growth cone to ultimately regulate β-actin local protein synthesis.

The m^6A/YTHDF1/APC pathway is essential for promoting axonal growth both in cultured hippocampal neurons and in callosal projection neurons *in vivo*

To determine the role of the m^6A/YTHDF1/APC pathway in axonal development, we transfected DIV2 hippocampal neurons with shRNAs targeting *Ythdf1*, *Mettl14*, or *Apc*, alongside GFP, and fixed them at DIV6. In comparison to neurons transfected with shControl, those expressing shRNAs against *Ythdf1*, *Mettl14*, and *Apc* exhibited a severe impairment in axonal growth, characterized by reduced primary branch length and total axonal branches (Figures 5A and 5B). The similar phenotypes resulting from m^6A pathway and APC deregulation suggest their involvement in a common molecular mechanism regulating axon development. To confirm that reduced APC protein levels drive the axonal defects caused by YTHDF1 knockdown, we co-transfected neurons with an APC-expressing plasmid alongside *shYthdf1* or *shControl*. Restoring APC expression in YTHDF1-knockdown neurons mitigated the axonal defects, primarily compensating for the impairment in axonal branching (Figures 5C and 5D). Next, we investigated whether this role in axonal development is conserved *in vivo*. First, we assessed the expression of YTHDF1 in the mouse neonatal brain at postnatal day (P)1, a critical period for callosal neurons to project axons interhemispherically through the corpus callosum. We performed immunostainings of YTHDF1, with an antibody validated on brain slices of a YTHDF1-knockout mouse model (data not shown), on coronal brain slices counterstained with DAPI, and we observed that YTHDF1 is widely expressed across the mouse brain, with an enrichment in the various layers of the cerebral cortex and the hippocampus (Figure S8A). To assess axonal projections, we performed *in utero* electroporation of tamoxifen-inducible *shYthdf1* constructs into the lateral ventricles of embryonic day (E)14 mouse embryos (Figures 5E and 5F). Pregnant mothers received tamoxifen at E17 and E18 to sustain YTHDF1 expression during neuronal proliferation and migration (Figure 5F). Cre recombinase expression was validated in electroporated cortical neurons (Figure S8B). By analyzing the length of the axons projecting to the corpus callosum, we found that downregulation of YTHDF1 led to a decrease in the length of the axons compared to the neurons electroporated with shControl (Figures 5G and 5H), indicating that YTHDF1 is required for axonal growth *in vivo*. Together, these findings highlight the essential role of the m^6A/YTHDF1/APC pathway in axon development, both in cultured neurons and in cortical projection neurons during brain development.

Autism- and schizophrenia-associated mutations in METTL14 compromise APC expression and axon development

To assess the relevance of the m^6A/YTHDF1/APC pathway in human disease, we examined the Human Gene Mutation Database and identified two METTL14 missense mutations linked

Figure 4. Deregulation of the m^6A/YTHDF1/APC molecular pathway causes defects in β-actin mRNA axonal localization and local translation

(A) Representative kymographs showing live recording of β-actin mRNA particles (red) in the axon of DIV6 neurons transfected with the β-actin mRNA reporter and either shControl or sh*Ythdf1*. The *x* axis corresponds to the position of the particles (μm), and the *y* axis to the time (seconds, from top to bottom). Scale bar, 10 μm.

(B) Quantification of the average speed, run length, instantaneous speed (number of granules: shControl, $n = 91$; sh*Ythdf1*, $n = 52$), and β-actin mRNA granule density (number of axons: shControl, $n = 30$; sh*Ythdf1*, $n = 21$).

(C) Representative images of axons from neurons transfected with shControl or sh*Ythdf1* along with a GFP-expressing plasmid (green), labeled with an anti-APC antibody (cyan) and β-actin ViewRNA fluorescence in situ hybridization (FISH) (red). The arrowheads indicate the co-distribution of APC and β-actin mRNA. Scale bar, 10 μm.

(D) Quantification of the number of β-actin mRNA and APC colocalizing puncta per μm in the axon (left) and the percentage of β-actin mRNA and APC colocalizing puncta relative to the total number of β-actin mRNA puncta (right) in the axon.

(E and F) Representative images of the puro-PLA newly synthesized β-actin signal (purple) in the axons (E) and growth cones (F) of shControl, sh*Ythdf1*, sh*Mettl14*, shControl + APC-HA, sh*Ythdf1* + APC-HA, and sh*Mettl14* + APC-HA DIV6 neurons. The arrowheads indicate the newly synthesized β-actin in the transfected neurons. Scale bars, 5 μm.

(G) Quantification of the number of β-actin puro-PLA puncta in the axons and growth cones of the conditions depicted in (E) and (F) (number of growth cones: ShControl, $n = 37$; sh*Ythdf1*, $n = 25$; sh*Mettl14*, $n = 35$; shControl + APC, $n = 65$; sh*Ythdf1* + APC, $n = 38$; sh*Mettl14* + APC, $n = 50$; number of axons: ShControl, $n = 36$; sh*Ythdf1*, $n = 23$; sh*Mettl14*, $n = 32$; shControl + APC, $n = 40$; sh*Ythdf1* + APC, $n = 30$; sh*Mettl14* + APC, $n = 30$).

The graphs represent the mean ± SEM in (D), (G), and (B) for β-actin mRNA density and mean ± SD in (B). $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, and $^{****}p < 0.0001$ or ns (not significant) comparing conditions to each other using one-way ANOVA tests (G) and unpaired t test (B and D). See also Figures S5, S6, and S7.

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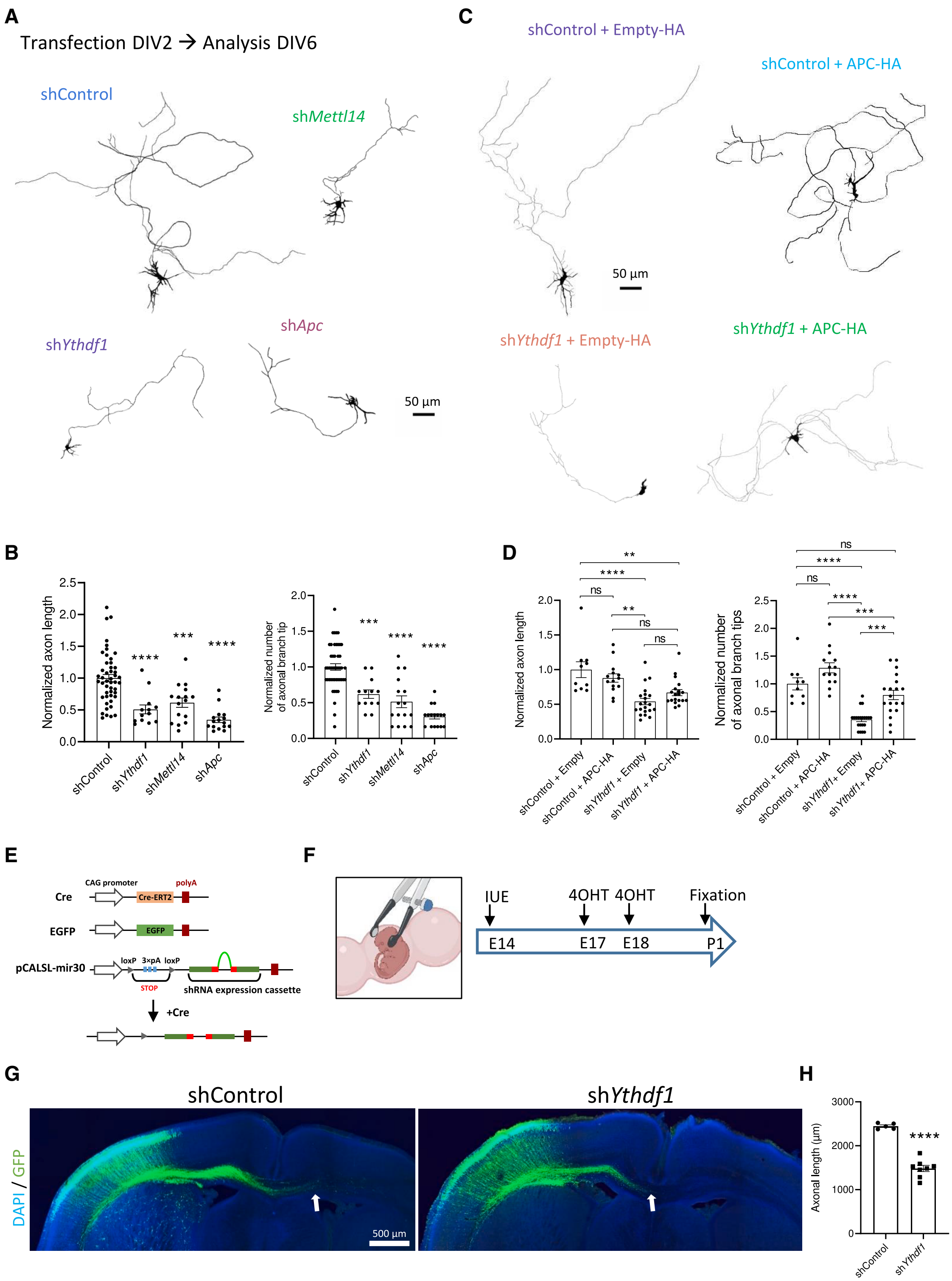

Figure 5. The m⁶A/YTHDF1/APC pathway is required for axonal growth in cultured neurons and callosal neurons *in vivo*

(A) Representative DIV6 neurons expressing shControl, sh*Ythdf1*, sh*Mettl14*, or sh*Apc* along with GFP. Scale bar, 50 µm.

(B) Quantification of the longest axonal branch length (number of axons: shControl, $n = 48$; sh*Ythdf1*, $n = 13$; sh*Mettl14*, $n = 16$; sh*Apc*, $n = 16$) and the total number of axonal branch tips (shControl, $n = 47$; sh*Ythdf1*, $n = 13$; sh*Mettl14*; $n = 16$; sh*Apc*, $n = 16$). The results are normalized to the average of the shControl.

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to neurological disorders, I311V (autism) and S399L (schizophrenia),[68–70] with unknown cellular effects. This is notable since *APC* is a risk gene for autism and schizophrenia, with its loss of function linked to these conditions.[16–25] These findings suggest that APC dysregulation and m^{6}A machinery dysfunction may contribute to a shared pathological pathway. To characterize these mutations, we mapped the affected residues onto the METTL3/METTL14 methyltransferase complex. Both I311 and S399 are at the METTL3-METTL14 interface (Figure S9A). Notably, D312, adjacent to I311, is crucial for methylation activity.[71] Simulated modeling showed that the autism-linked I311V mutation reduces the number of residues in METTL3 with hydrophobic interactions (Figure S9B). S399, a phosphorylated residue forming a salt bridge with R471 in METTL3, is disrupted in the schizophrenia-associated S399L mutation, eliminating its negative charge. To assess their impact, we introduced these mutations into mouse METTL14 (99% identical to human) and validated their expression in neurons and Neuro2a cells (Figures S9C–S9E). Like wild type (WT), the mutants localized mainly in the nucleus (Figure S9C). Co-immunoprecipitation confirmed that both variants interact with METTL3 at normal levels, preserving complex formation (Figure S9F). Then, we performed RIP-qPCR and showed that overexpressing these mutants significantly reduced YTHDF1 binding to *Apc* mRNA (Figures 6A and 6B). These findings suggest that mutated METTL14 variants form a complex with METTL3, possibly compromising its methyltransferase activity, ultimately disrupting YTHDF1 binding to *Apc* mRNA and its translation. Next, we overexpressed the mutants in cultured neurons and analyzed APC expression, growth cone size, and axon development. Both I311V and S399L, but not WT, reduced APC fluorescence in the soma and neurites at DIV6 (Figures 6C–6E). Growth cone area was also diminished (Figures 6F and 6G). Furthermore, neurons expressing these variants showed shorter primary axons and fewer branches (Figures 6H and 6I). These findings indicate that schizophrenia- and autism-linked METTL14 mutations reduce APC protein level and impair axon development, suggesting that m^{6}A/YTHDF1/APC pathway dysregulation contributes to these neurological disorders.

DISCUSSION

RBPs are essential *trans*-acting factors that govern every stage of the mRNA life cycle, including mRNA splicing, trafficking, stability, and translation, thereby playing a significant role in neurodevelopment and synaptic function.[72] Here, we demonstrate that the m^{6}A machinery is involved in the control of translation of an RBP, APC, which is crucial for the regulation of the axonal cytoskeleton dynamics at the growth cone by mediating the axonal transport and local translation of β-actin mRNA (Figure 7).

Two recent studies examining the neuronal subcellular localization of selected mRNAs have seemingly reached contradictory findings regarding the role of m^{6}A in this process. In the first study, the authors showed that m^{6}A residues in the 3′UTR promote the localization of specific neuronal mRNAs to dendrites and axons, with profiling revealing thousands of enriched m^{6}A sites in neurites compared to cell bodies.[56] They also show that depletion of METTL3 alters the localization of various transcripts, including *Camk2a* and *Map2*, which require m^{6}A residues for neurite localization, suggesting a role for m^{6}A in neuronal mRNA localization. Conversely, in the second study, the authors demonstrated that m^{6}A-enriched transcripts are biased toward the cell soma versus axons as a consequence of lower stability.[57] Our results introduce a new model in which the m^{6}A-mediated control of APC protein translation in the cell soma could regulate the transport and local translation of axonal mRNAs related to APC RNP granules, regardless of their m^{6}A-modification status. APC binds to 260 mRNAs that primarily encode cytoskeletal proteins related to APC functions.[8] A significant portion of APC-bound mRNAs (70%) intersect with our previously published m^{6}A-mRNAs,[8,40] which indicates that the transport and local translation of those as well as the remaining 30% of APC-bound and m^{6}A-devoid mRNAs could be impacted by the deregulation of APC protein translation upon METTL14 or YTHDF1 deregulation. While our study focuses mainly on how the m^{6}A/YTHDF1/APC pathway modulates the transport and translation of β-actin but not *Tubb2b* mRNA, both APC's target mRNAs, it would be interesting in the future to investigate whether other APC-related mRNAs are similarly deregulated and whether their m^{6}A-modification status is relevant. A recent study using an *in vitro* reconstitution system found that APC and FMRP bind *Tubb2b* mRNA with comparable affinity, whereas FMRP shows much weaker binding to β-actin mRNA compared to APC.[67] As we found that YTHDF1 knockdown does not impact FMRP protein level (Figure S7), this raises the possibility that FMRP could compensate for *Tubb2b* mRNA transport but not for β-actin in the absence of YTHDF1. It is interesting to note that Flamand and Meyer observed that YTHDF1 depletion in neurons did not affect the transport of *Camk2a* and *Map2* mRNAs.[56] Given that *Camk2a* and *Map2* mRNAs are not APC's target mRNAs,[56] these data are in accordance with our findings and suggest that YTHDF1 is involved in the specific control of the transport and translation of select mRNAs through the regulation of APC protein expression level. Is the expression of other RBPs also controlled by the m^{6}A machinery through a similar mechanism? FMRP and TDP-43 are known to

(C and D) Representative DIV6 neurons (C) and quantifications (D) of axon length and branching after re-expressing APC in YTHDF1-knockdown neurons (number of axons: shControl + empty, $n = 10$; shControl + APC-HA, $n = 14$; sh*Ythdf1* + empty, $n = 21$; sh*Ythdf1* + APC-HA, $n = 19$). The results are normalized to the average of the shControl + empty. Scale bar, 50 μm.

(E) Plasmids used for the Cre-inducible expression of shRNA in callosal neurons.

(F) The experimental scheme of *in utero* electroporation (IUE) (E14), tamoxifen (4OHT) injections (E17 and E18), and brain tissue collection at P1.

(G) Representative images of P1 brain slices showing the electroporated callosal neurons expressing either shControl or sh*Ythdf1* along with GFP. The arrows indicate the tip of the contralaterally projecting axon bundles. Scale bar, 500 μm.

(H) Quantification of the maximum axonal length of electroporated callosal neurons (number of brains: shControl, $n = 5$; shYthdf1, $n = 8$).

The graphs represent the mean ± SEM in (B), (D), and (H). *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, and ****$p < 0.0001$ or ns (not significant) comparing conditions to each other using one-way ANOVA tests (B and D) and unpaired t test (H). See also Figure S8.

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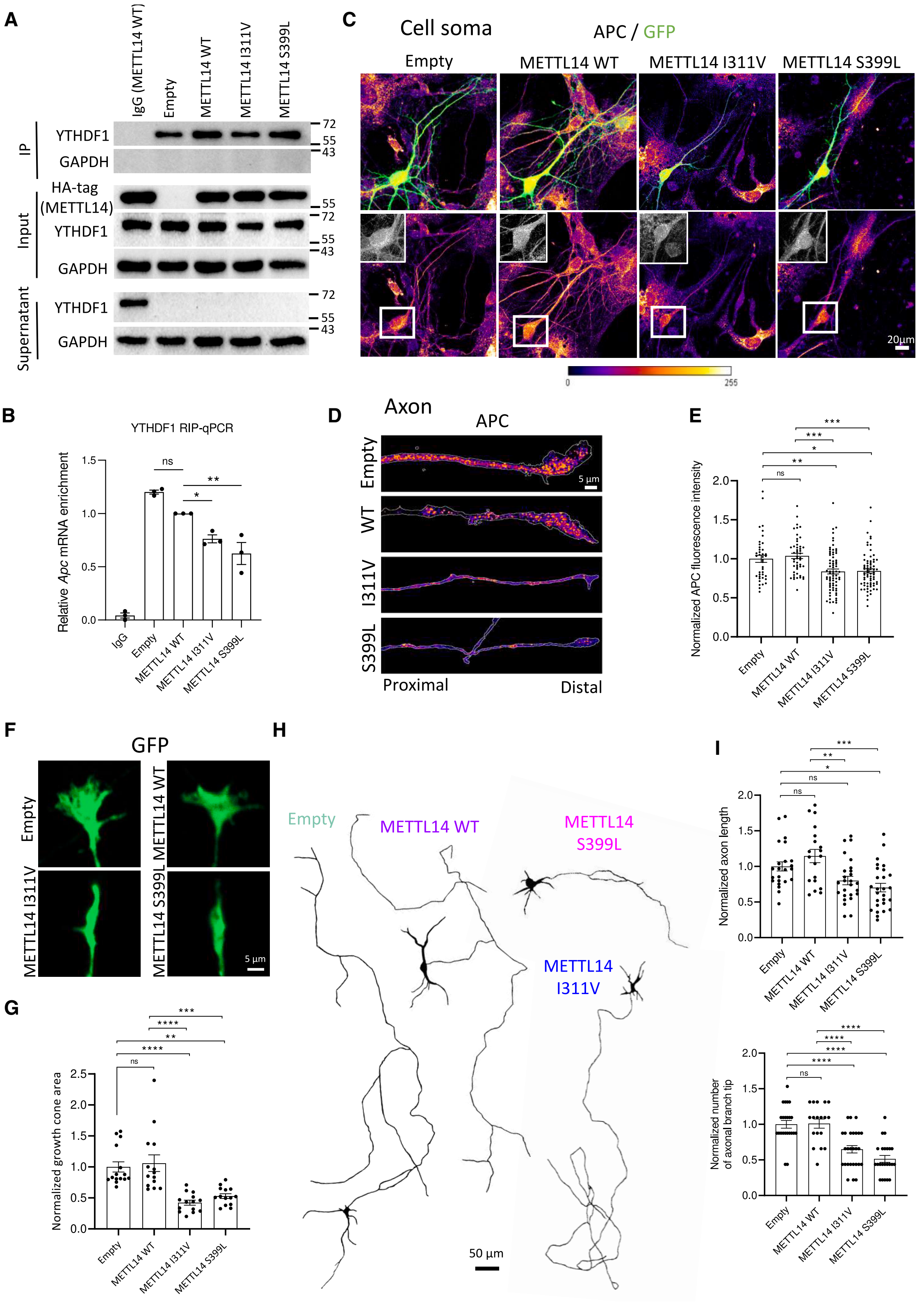

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be involved in mRNA transport and local translation in axons. Our data showed that the number of FMRP granules in axons or dendrites does not change after YTHDF1 knockdown, suggesting that m^{6}A control of APC expression is not applied as a general mechanism for localized RBPs in neurons (Figure S7).

Despite the known crucial roles of the microtubule and actin cytoskeleton in maintaining neuronal functions, no connection has been established between m^{6}A and the cytoskeleton to date. In our study, we revealed that YTHDF1 deregulation leads to altered microtubule and actin cytoskeleton dynamics in the growth cone and that the decrease in F-actin is likely to be due, at least partially, to a reduction in β-actin local translation. Our data indicating that APC reexpression after YTHDF1 depletion restores β-actin local translation and compensates for axon branching defects further confirm the well-established role of locally synthesized β-actin in guiding axon growth and branching during neuronal development.[73–78] It is important to mention that APC was recently described as being crucial for the local assembly of branched actin filaments in the growth cone of hippocampal neurons.[65] APC targets microtubule tips and actin-branched regions within the growth cone to stimulate the assembly of branched actin networks on microtubule plus-ends, and this function is likely to depend on the ability of APC to stimulate the Arp2/3 signaling pathway.[65,79] It therefore remains to be delineated in future studies the specific contributions of the regulation of β-actin local translation by APC compared to its role as an microtubule plus-end actin branching nucleator in the maintenance of axonal growth and branching. As local translation only provides with a small amount of β-actin in comparison to the existing pool of long-lived β-actin, it is tempting to postulate that APC may additionally serve as a regulator of growth cone maintenance by coordinating local translation and the subsequent incorporation of newly synthesized β-actin at the interface between microtubule plus-ends and the branched actin network.

m^{6}A methylation and YTHDF readers have been implicated in the regulation of axon growth and guidance in various neuronal models and developmental stages. In particular, YTHDF1 regulates the translation of Robo3.1, a protein crucial for guiding pre-crossing axons in the embryonic spinal cord.[35] Additionally, YTHDF1 and YTHDF2 are closely associated with axonal growth in cerebellar granule cells as the depletion of YTHDF1 and YTHDF2 enhances axonal growth in granule cells by modulating the local translation of components involved in Wnt5a signaling.[36] Furthermore, in *Drosophila*, Ythdf (there is only one ortholog in *Drosophila*) plays a role in limiting axonal growth at larvae neuromuscular junctions by modulating the translation of transcripts involved in axonal growth.[37] Our results showing that YTHDF1 knockdown reduces axon growth and branching in primary neuron cultures and in developing callosal projection neurons appear to be in contradiction with the findings in granule cells and the *Drosophila* larvae neuromuscular junction. However, our results align with a recent work demonstrating that downregulation of the long non-coding Dppa2 upstream binding RNA (*Dubr*) leads to the degradation of YTHDF1 and YTHDF3, resulting in delayed axonal growth in cortical neurons.[38] Interestingly, our findings are also consistent with the previously published embryonic and postnatal axonal translatome datasets indicating that the translation of APC-related mRNAs is most pronounced in the youngest axons (E17.5) and gradually declines in postnatal stages.[80] This implies that YTHDF1, as well as the other YTHDFs, likely plays a critical role in regulating axonal growth during axonal development through the modulation of APC translation but that this function may be context dependent, varying between young and adult axons, neuronal populations, and species.

Finally, given that *APC* has been identified as a risk gene for autism spectrum disorder and schizophrenia,[16,17,20–23,25] our findings suggest that deregulation of the m^{6}A/YTHDFs/APC axis pathway could contribute to the physiopathology of these neurological disorders. Supporting this notion, we provide evidence that exogenous expression of METTL14 variants, associated with autism and schizophrenia,[68–70] results in reduced APC expression and axon growth defects. This further suggests that dysregulation of the m^{6}A/YTHDFs/APC pathway may contribute to a shared pathological mechanism underlying these disorders. The implication of m^{6}A in neurological disorders opens new therapeutic possibilities as CRISPR-Cas13-based m^{6}A editing tools have emerged as a promising strategy to modify specific mRNA targets.[81,82] Furthermore, our findings provide insights into the regulation of local translation in neurons, a key process implicated in neurological diseases associated with translation

Figure 6. Autism- and schizophrenia-associated mutations in METTL14 disrupt APC protein expression and impair axon development

(A and B) RIP-qPCR analysis of *Apc* mRNA bound to YTHDF1; the immunoblots in (A) show the efficiency of YTHDF1 IP in IgG, empty vector, METTL14 WT, METTL14 I311V, and S399L; the bar graph in (B) shows the *Apc* levels bound to YTHDF1 in these conditions. *Apc* mRNA expression is relative to METTL14 WT, and *Apc* mRNA abundance was normalized to *Gapdh* in input RNA, $n = 3$.

(C and D) Representative images of DIV6 cell soma (C) and axons (D) from neurons transfected with empty vector, METTL14 WT, I311V, or S399L, along with GFP, labeled with an anti-APC antibody (Fire LUT). The insets show a higher magnification of APC staining in the cell soma of transfected neurons. Scale bar, 20 μm in (C) and 5 μm in (D).

(E) Quantification of APC fluorescence intensity in the cell soma and neurites of neurons transfected with empty vector, METTL14 WT, I311V, or S399L constructs (number of neurons: empty vector, $n = 40$; METTL14 WT $n = 48$; I311V, $n = 73$; S399L $n = 70$). The results are normalized to the average of the empty control.

(F) Representative images showing the axonal growth cones of DIV6 neurons transfected with a GFP-expressing plasmid and the empty vector, METTL14 WT, I311V or S399L at DIV2. Scale bar, 5 μm.

(G) Quantification of the growth cone area (number of growth cones: empty vector, $n = 15$; METTL14 WT $n = 14$; I311V, $n = 14$; S399L $n = 14$). The results are normalized to the average of the empty control.

(H) Representative DIV6 neurons expressing empty vector, METTL14 WT, I311V, or S399L constructs along with GFP. Scale bar, 50 μm.

(I) Quantification of the longest axonal branch length (number of axons: empty vector, $n = 23$; METTL14 WT $n = 19$; I311V, $n = 26$; S399L $n = 27$) and the total number of axonal branch tips (empty vector, $n = 23$; METTL14 WT $n = 18$; I311V, $n = 27$; S399L $n = 23$). The results are normalized to the average of the empty control.

The graphs represent the mean ± SEM in (B), (E), (G), and (I). *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, and ****$p < 0.0001$ or ns (not significant) comparing conditions to each other using one-way ANOVA tests (B, E, G, and I). See also Figure S9.

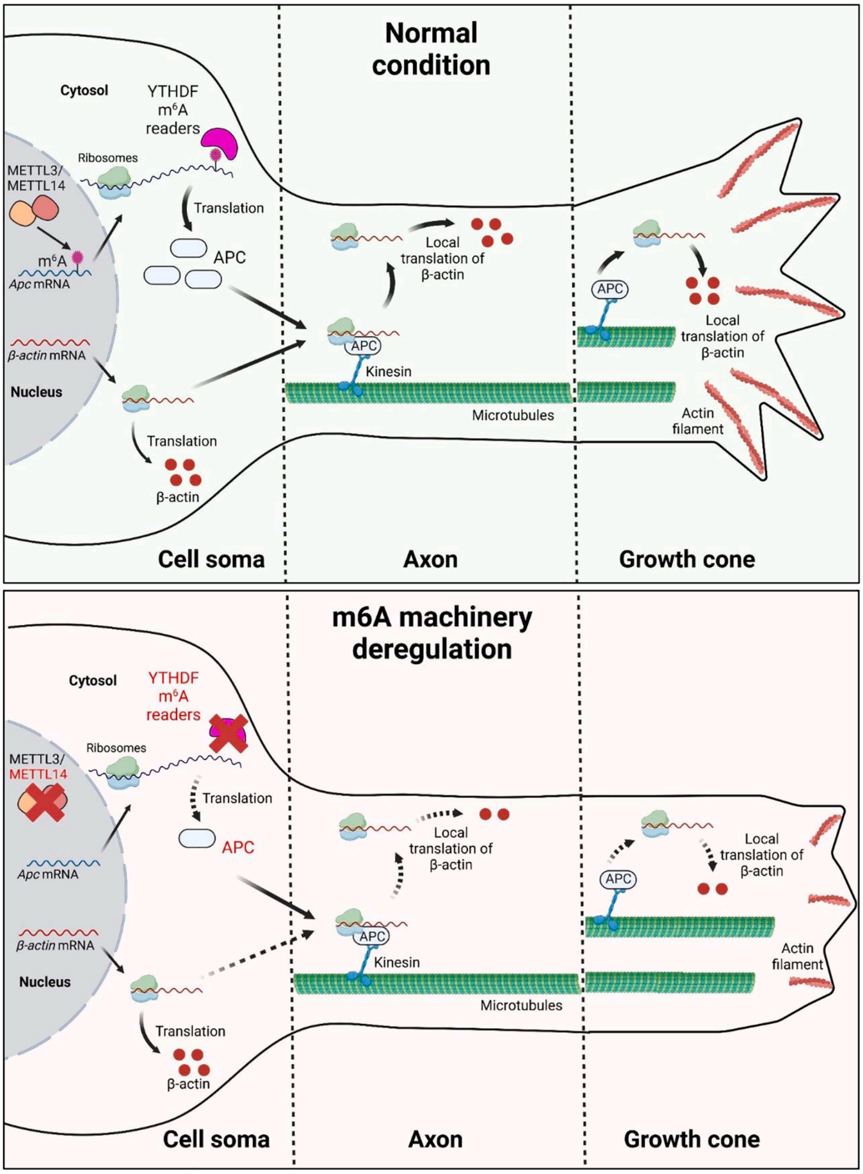

Figure 7. Model depicting the regulation of axon development via the m^{6}A/YTHDF1/APC pathway, which governs β-actin mRNA transport and local translation

defects, including fragile X syndrome, amyotrophic lateral sclerosis, and frontotemporal dementia.

Limitations of the study

Although we identified bona fide m^{6}A-modification sites on *Apc* mRNA using the SELECT method, technical limitations prevented us from directly linking these sites to observed axon defects. Future research will focus on identifying the specific functions of m^{6}A sites on *Apc* mRNA and APC protein translation. This could be achieved using CRISPR-based strategies to selectively methylate or demethylate these sites. Our study reveals that global APC protein expression decreases when YTHDF readers or METTL14 are downregulated and that it is accompanied by a reduction in β-actin axonal local protein synthesis. Beyond its role in β-actin local translation, APC also regulates actin filaments through its scaffolding functions. Future investigations should distinguish APC's role in β-actin local translation from its impact on cytoskeletal dynamics as a scaffold protein. Here, we employed *in utero* electroporation of shRNA to demonstrate YTHDF1's role in axon development during embryonic/early postnatal development. However, our study has limitations, as the long-term effects of axonal defects caused by YTHDF1 knockdown remain unknown. Additionally, the precise mechanisms by which YTHDF m^{6}A readers interact to regulate *Apc* mRNA translation require further exploration.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Dan Ohtan Wang (ohtan.wang@nyu.edu).

Materials availability

All materials generated in this study will be shared by the lead contact upon request.

Data and code availability

- This study does not report original code.
- All data reported in this paper will be shared by the lead contact upon request.
- Any additional information required to reanalyze the data reported in this work is available from the lead contact upon request.

ACKNOWLEDGMENTS

L.B.'s postdoctoral fellowship was supported by the Japan Society for the Promotion of Science (JSPS; award number: P21085). R.R. was supported by the Sasakawa Research Grant (award number: 2018-4027), Mitsubishi Corporation International Scholarship (award number: MITSU1824), and JASSO Honor Fellowship. This work is supported by KAKENHI (award numbers: 22KF0399 and 21H02580) from Japan and Faculty Research Award from New York University Abu Dhabi to D.O.W. The graphical abstract and Figure 7 were created with BioRender.com. We thank Drs. Takahiro Fujiwara and Fumiyoshi Ishidate at iCeMS Imaging Center and Dr. Yoshie Fujiwara for their technical support, and we thank NYUAD Center for Brain and Health for helpful discussions.

AUTHOR CONTRIBUTIONS

L.B., R.R., and D.O.W. conceived and designed the project. L.B. designed, performed, and analyzed the experiments and wrote the manuscript. R.R. and B.S. performed the experiments and contributed to manuscript writing. I.O. and H.U. performed part of the experiments. D.O.W. edited the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.celrep.2025.115727.

Received: April 29, 2024
Revised: January 14, 2025
Accepted: April 30, 2025
Published: May 20, 2025

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti-YTHDF1	Proteintech	Cat#:17479-1-AP; RRID:AB_2217473
Anti -YTHDF2	Proteintech	Cat#:24744-1-AP; RRID:AB_2687435
Anti-YTHDF3	Santa Cruz Biotech	Cat#:Sc-377119; RRID:AB_2687436
Anti-METTL14	ATLAS ANTIBODIES (Sigma)	Cat#:HPA038002; RRID:AB_10672401
Anti-Tubulin (Tyrosinated)	Abcam	Cat#:ab6160; RRID:AB_305328
Anti-APC	Millipore	Cat#:OP80; RRID:AB_2057371
Anti-APC-C (C-20)	Santa Cruz	Cat#:SC-896; RRID:AB_2057493
Anti-FMRP	Abcam	Cat#:ab17722; RRID:AB_2278530
Anti-HA-tag	MBL	Cat#:M180-3; RRID:AB_10951811
Anti-HA tag	Abcam	Cat#:ab9110; RRID:AB_307019
Anti-GAPDH (14C10)	Cell Signaling	Cat#:2118; RRID:AB_561053
Anti-Puromycin	Millipore	Cat#:MABE343; RRID:AB_2566826
Anti-DCX	Abcam	Cat#:ab18723; RRID:AB_732011
Anti-CRMP2	Biolegend	Cat#:869001; RRID:AB_732011
Anti-GFP	Abcam	Cat#:ab13970; RRID:AB_300798
Anti-Tau	Abcam	Cat#:ab75714; RRID:AB_1310734
Anti-MAP2	Abcam	Cat#:ab5392; RRID:AB_2138153
Anti-Beta-actin	Cayman Chemicals	Cat#:32127; RRID:AB_2810170
Anti-Cre Recombinase	Millipore	Cat#:MAB3120; RRID:AB_2085748
Anti-Tubb2b	Abcepta	Cat#:AP11940a; RRID:AB_10820514
Goat Anti-Mouse IgG (H + L) cross-adsorbed secondary antibody Alexa Fluor 488	Thermofisher	Cat#:A11001; RRID:AB_2534069
Goat Anti-Mouse IgG (H + L) cross-adsorbed secondary antibody Alexa Fluor 568	Thermofisher	Cat#:A11004; RRID:AB_2534072
Goat Anti-Mouse IgG (H + L) cross-adsorbed secondary antibody Alexa Fluor 633	Thermofisher	Cat#:A21050; RRID:AB_2535718
Goat Anti-Rabbit IgG (H + L) cross-adsorbed secondary antibody Alexa Fluor 633	Thermofisher	Cat#:A21070; RRID:AB_2535731
Donkey anti-Chicken IgY (H + L) Highly Cross Adsorbed Secondary Antibody, Alexa Fluo 568	Thermofisher	Cat# A78950; RRID:AB_2921072
Bacterial and virus strains		
NEB Stable Competent E. coli	New England Biolabs	Cat#:C3040H
Chemicals, peptides, and recombinant proteins		
Puromycin	InvivoGen	Cat#:58-58-2
ProLong Diamond antifade mountant	Thermofisher	Cat#:P36965
DPBS	Thermofisher	Cat#:10010023
Neurobasal medium	Thermofisher	Cat#:10888022
MEM	Thermofisher	Cat#:11095080
Penicillin/Streptomycin	Thermofisher	Cat#:15140122

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Lipofectamine 2000	Thermofisher	Cat#:1668019
Poly-L-lysine	Merck	Cat#:P6282
Horse serum	Thermofisher	Cat#:16050122
D-glucose	Thermofisher	Cat#:15384895
Sodium Pyruvate (100 mM)	Thermofisher	Cat#:11360070
B-27 Supplement (50X), serum free	Thermofisher	Cat#:17504044
GlutaMAX Supplement	Thermofisher	Cat#:35050061
Donkey serum	Merck	Cat#:D9663
DAPI	Invitrogen	Cat#:62248
All Trans Retinoic Acid (ATRA)	Sigma	Cat#:R2625
Alexa Fluor 488 Phalloidin	Thermofisher	Cat#:A12379
Alexa Fluor 594 Phalloidin	Thermofisher	Cat#:A12381
Fetal bovine serum	Thermofisher	Cat#:10270106
4%-Paraformaldehyde	Nacalai Tesque	Cat#:09154-85
cOmplete, Mini, EDTA-free Protease Inhibitor Cocktail	Merck	Cat#:11836170001
PVDF membrane	Bio-Rad	CAT#:1620177
dNTP	New England Biolabs	Cat#:N0447
CutSmart buffer	New England Biolabs	Cat#:B6004S
Bst 2.0 DNA polymerase	New England Biolabs	Cat#:M0537S
SplintR ligase	New England Biolabs	Cat#:M0375S
Fast green	Merck	Cat#:F7258
4-Hydroxytamoxifen	Merck	Cat#:H6278
Progesterone	Merck	Cat#:P0130
Corn oil	Merck	Cat#:C8267
Low melting agarose	Bio-Rad	CAT#:1613111
TRIzol Reagent	ThermoFisher	CAT#:15596026
Critical commercial assays		
ViewRNA Cell Plus Assay kit	ThermoFisher	Cat#:88-19000-99
ViewRNA Cell Plus Probe Sets *Actb* mRNA	ThermoFisher	Cat#:VB6-12823-VC
ViewRNA Cell Plus Probe Sets *Apc* mRNA	ThermoFisher	Cat#: VB6-19724-VC
EndoFree Plasmid Maxi Kit	Qiagen	Cat#:12362
RNeasy Mini Kit	Qiagen	Cat#:74104
ECL Select™ Western Blotting Detection Reagent	Merck	Cat#:GERPN2235
iScript RT Supermix	Bio-Rad	Cat#:1708840
PowerUp SYBR Green Master Mix	Thermofisher	Cat#:A25742
Neuron Dissociation Solutions	Fujifilm	Cat#:291-78001
Pierce™ BCA Protein Assay Kit	Thermofisher	Cat#:23225
Direct-zol RNA MiniPrep Kit	Zymo Research	Cat#:R2050
SuperScript™ III Reverse Transcriptase	ThermoFisher	Cat#:18080093
NZYSupreme qPCR Green Master Mix (2x), ROX	Nordic Biosite	Cat#:MB44101
Dynabeads™ Protein G for Immunoprecipitation	Thermofisher	Cat#:10003D
Duolink *In Situ* Detection Reagents FarRed	Merck	Cat#:DUO92013
Duolink *In Situ* PLA Probe Anti-Rabbit PLUS	Merck	Cat#:DUO92002
Duolink *In Situ* PLA Probe Anti-Mouse MINUS	Merck	Cat#:DUO92004

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Experimental models: Cell lines		
Neuro-2a neuroblastoma cells	ATCC CCL-131	RRID:CVCL_0470
Experimental models: Organisms/strains		
ICR mice	Japan SLC	RRID:MGI:2160272
Oligonucleotides		
SELECT_qPCR_F 5′ATGCAGCGACTCAGCCTCTG3′	FASMAC	N/A
SELECT_qPCR_R 5′TAGCCAGTACCGTAGTGCGTG3′	FASMAC	N/A
APC m^6A 7836 up 5′tagccagtaccgtagtgcgtg GAAGGGAAGGGCTTAAAACTGGGG3’	FASMAC	N/A
APC m^6A 7836 down 5phos/CTGTGCC TGCGACCTGGTcagaggctgagtcgctgcat3’	FASMAC	N/A
APC m^6A 11110 up 5′tagccagtaccgtagt gcgtgCAGCCATTATCTCTAGCCAAAGGG3’	FASMAC	N/A
APC m^6A 11110 down 5phos/TTAATCCAGCA ACTCCATCTTCAAGGcagaggctgagtcgctgcat3’	FASMAC	N/A
APC m^6A 12324 up 5′tagccagtaccgtagtgc gtgGAACAGGAGGAAAATGACCTGGG3’	FASMAC	N/A
APC m^6A 12324 down 5phos/TTACAGAAGTCA CGGTTCATAGAATTCCcagaggctgagtcgctgcat3’	FASMAC	N/A
APC A 11121 up 5′tagccagtaccgtagtgcgtg CCCTTTGCTTTCACAGCCATTATCTC3’	FASMAC	N/A
APC A 11121 down 5phos/AGCCAAAGGGTTT AATCCAGCAACcagaggctgagtcgctgcat3’	FASMAC	N/A
Ythdf1 forward qPCR 5′GCATCAGAAGGATGCAGTTCATG3′	FASMAC	N/A
Ythdf1 Reverse qPCR 5′GATGGTGGATAGTAACTGGACAG3′	FASMAC	N/A
Mettl14 forward qPCR 5′AGAGTGCGGATAGCATTGGTGC3′	FASMAC	N/A
Mettl14 Reverse qPCR 5′CTCCTTCATCCAGACACTTCCG3′	FASMAC	N/A
Apc forward qPCR 5′CCCCAGTGACCTTCCAGATA3′	FASMAC	N/A
Apc reverse qPCR 5′GCACTGTCTGTGGAGGAGGT3′	FASMAC	N/A
sh*Ythdf1* 5′-CTAATACATCCCAAAGATA-3′	FASMAC	N/A
sh*Ythdf2* 5′-GGATTAATTTGATTTCAAA-3′	FASMAC	N/A
sh*Ythdf3* 5′-GCCAAACAGTACATAATCTA-3′	FASMAC	N/A
sh*Mettl14* 5′-CTTACACTTGCATCTTTAT-3′	FASMAC	N/A
sh*Apc* 5′-GTACTAACTAGGTAAATA-3′	FASMAC	N/A
shControl 5′-GCTACCTCCATTTAGTGT-3′	FASMAC	N/A
Recombinant DNA		
pCALSL-mir30	Addgene	#13786
pCAG-ERT2CreERT2	Addgene	#13777
pCAGGS-HA	Gift from Kengaku Mineko	N/A

(*Continued on next page*)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
pHR-tdPP7-3xmCherry	Addgene	#74926
pcDNA4TO-24xGCN4_v4-kif18b-24xPP7	Addgene	#74928
5′UTR-SunTag24x-Actin-3′UTR-PP724x	This study	N/A
pCAG-EGFP	Gift from Kengaku Mineko	N/A
pSUPER vector	Gift from Kengaku Mineko	N/A
pCAGGS-HA-APC	This study	N/A
Software and algorithms		
Fiji (ImageJ)	NIH, USA	https://fiji.sc/; RRID: SCR_002285
GraphPad Prism 8	GraphPad Software	https://www.graphpad.com/scientificsoftware/prism/; RRID: SCR_002798
ImageJ plugin KymoToolBox	fabrice.cordelieres@curie.u-psud.fr	N/A
StepOne Real-Time PCR Software v2.3	Applied Biosystems	RRID:SCR_023455
ZEN software	Zeiss	RRID:SCR_013672

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Animals

For primary culture of neurons, ICR male and female mice at postnatal day 0 were purchased from SLC or Shimizu (Japan). For *in utero* electroporation, pregnant mice were purchased from SLC or Shimizu and housed at the Institute for integrated cell-material Sciences (iCeMS) or RIKEN Center for Biosystems Dynamics Research (BDR) animal facilities. *In utero* electroporation was performed in embryonic day 14.5 embryos and electroporated male and female mice at postnatal day 1 were dissected. This study was carried out in accordance with the Guide for the Care and Use of Laboratory Animals from the Society for Neuroscience and was authorized by the Animal Care and Use Committee of Kyoto University and the RIKEN BDR.

In utero electroporation of mouse embryos

Timed-pregnant (embryonic day E14.5) mice were anesthetized with isoflurane before the injection of painkillers. Endotoxin-free plasmids mixed with 0.1% fast green (Sigma) were injected into lateral ventricles of the embryo forebrains, followed by electroporation (5 pulses of 40 mV at 50 ms intervals for 950 ms) using 3mm diameter platinum electrodes (Sonidel) connected to an electroporator (ECM 830, BTX). The electroporation was performed with a combination of three plasmids consisting of a tamoxifen-inducible CreERT2 expressing plasmid and a pCAGGS-EGFP plasmid along with either shControl or sh*Ythdf1*. The embryos were placed back in the abdominal cavity. For injection, 4OHT and progesterone were dissolved in 100% ethanol at concentrations of 20 mg/mL and 10 mg/mL, respectively, and then further diluted with nine volumes of corn oil (Sigma-Aldrich, Merck). The diluted tamoxifen solution (2 mg/mL, 100 μL per mouse) was administered intraperitoneally two times at E17 and E18 and the brains of the pups were dissected at P1 and fixed in 4% PFA overnight.

Primary culture of neurons and transfection

Hippocampi were dissected from postnatal day 0 (P0) mice. Hippocampal neurons were dissociated using neuron dissociation solutions (Wako) and plated on 12 mm coverslips or 35 mm glass-bottom dishes (MatTek) coated with poly-L-lysine (Sigma) in MEM (Thermofisher) supplemented with 10% horse serum (Thermofisher), 0.6% D-glucose (Thermofisher), 1 mM sodium pyruvate (Thermofisher) and 1% penicillin-streptomycin (Thermofisher). Three hours after plating, the media was replaced by growth media consisting of Neurobasal-A medium (Thermofisher) supplemented with B-27 supplement, GlutaMAX and penicillin-streptomycin. Cells were maintained at 37°C, 5% CO_2 until fixation or live-imaging.

Neuro2a cell line culture and transfection

Mouse Neuro2a neuroblastoma cells (RRID:CVCL_0470), obtained from ATCC (ATCC CCL-131), were authenticated by the supplier. Experiments were performed using cells at passages P5–P7. Cell morphology was regularly examined under a microscope to ensure cellular integrity, and mycoplasma contamination was routinely screened. The cells were cultured in Minimum Essential Media (MEM, 1X, Thermofisher) supplemented with 10% (v/v) fetal bovine serum (FBS, heat-inactivated, Thermofisher) without any antibiotic supplement. Neuro-2a cells were maintained at 37°C in 5% CO_2 until ready for downstream experiments. Neuro-2a cells were placed in MEM with 10% FBS without Pen-Strep 24h before transfection using Lipofectamine 2000 (Thermofisher). Media was changed 24 h after transfection with 10% Fetal Bovine Serum containing MEM media with Pen-Strep. Neuro-2a cells were subjected to differentiation 48h after transfection using MEM media containing low serum (0.5% FBS) with ATRA (final concentration 10 μM). Cells were maintained for one more day before harvesting for western blotting or qPCR analysis.

METHOD DETAILS

Plasmids and oligos

For the construction of knockdown vectors, synthetic oligos were designed and cloned into BglII/HindIII sites or BglII/XhoI sites on pSUPER vector protocol (Oligoengine), and into XhoI/EcoRI sites of pCALSL-mir30 (Addgene plasmid #13786). The oligos for the knockdown experiments mentioned below are designed against mouse genome: shControl: 5′-GCTACCTCCATTTAGTGT-3’; sh*Ythdf1*: 5′-CTAATACATCCCAAAGATA-3’; sh*Mettl14*: 5′-CTTACACTTGCATCTTTAT-3’; sh*Apc*: 5′-GTACTAACTAGGTAAATA-3’. pCAG-ERT2CreERT2 plasmid was purchased from Addgene (plasmid #13777). For SELECT experiments, mouse full-length cDNA of ALKBH5 was subcloned into a pCAGGS-HA vector. For β-actin mRNA transport analysis, pHR-tdPP7-3xmCherry and pcDNA4TO-24xGCN4_v4-kif18b-24xPP7 were purchased from Addgene (plasmids #74926 and #74928, respectively). Briefly, the CMV promoter was replaced by a CAG promoter using HindIII/MluI and the mus musculus β-actin mRNA CDS (GenBank: NM_007393.5) was inserted using NheI/AgeI. The 5′UTR was inserted using HindIII/HindIII and finally, the 3′UTR was inserted using BamHI/NheI to obtain the reporter plasmid 5′UTR-SunTag24x-Actin-3′UTR-PP724x. SunTag24x-Tubb2b-3′UTR-PP724x was generated by replacing β-actin mRNA CDS and 3′UTR with *Tubb2b* mRNA CDS and 3′UTR using BamHI/AgeI and the β-actin mRNA 5′UTR was removed with HindIII/HindIII. pCAG-EGFP plasmid was a kind gift from Kengaku Mineko’s lab (iCeMS, Kyoto University, Japan). Mus musculus *Apc* full length coding sequence, without UTRs, (GenBank: NM_001402731.1), was subcloned in a pCAGGS-HA vector using BamHI/EcoRI to express APC protein with an N-terminal HA tag.

Immunostaining in primary neuronal culture

Dissociated hippocampal cultures (DIV6) were fixed in 4% PFA/PBS and permeabilized in 0.1% Triton X-100/PBS for 10 min at RT. Cells were blocked in 10% donkey serum/PBS for at least 1 h before primary antibody incubation (in 10% donkey serum/PBS, overnight, 4°C). Next day, after 3 times washing with 1X PBS, followed by secondary antibody incubation in the dark for 1.5 h, samples were counterstained with DAPI (1 μg/mL in PBS for 10 min at RT) and mounted onto glass slides with ProLong Diamond Antifade Mountant (ThermoFisher). To label F-actin in axonal growth cones, dissociated hippocampal cultures (DIV6) were fixed in 4% PFA (20 min at 37°C), 3 times washed with 1X PBS and permeabilized in 0.1% Triton X-100/PBS for 10 min at RT. For single staining, Phalloidin conjugated with fluorophore (488 or 594) solution was diluted at 1:1000 to 1:500 using the dilution buffer composed of 1X PBS and 1% BSA. The neurons with the Phalloidin solution were incubated for 90 min. After 3 times washing with 1X PBS, followed by MilliQ wash, samples were mounted onto glass slides with ProLong Diamond Antifade Mountant (ThermoFisher). Images of tyrosinated-tubulin and F-actin stainings in the growth cone were acquired using a confocal microscope system Dragonfly 500 (Oxford Instruments).

Immunostaining on brain sections

Fixed *in utero* electroporated brains at postnatal day 1 were embedded in agarose 4% low melting agarose (Bio-Rad) and cut into coronal sections (100 μm) using a vibratome (Leica VT1000S, Leica Microsystems). For immunostainings coronal sections were permeabilized in 0.3% Triton X-100/PBS, and blocked in 10% donkey serum in 1X PBS for 1h. Primary antibodies were incubated overnight at 4°C with 0.3% Triton X-100/PBS and 1% donkey serum. The following antibodies were used: anti-Cre Recombinase (MAB3120, 1:5000, Millipore); anti-YTHDF1 (17479-1-AP, 1:200, Proteintech); anti-GFP (ab13970, 1:2000, Abcam). After washing in 0.3% Triton X-100/PBS, the sections were incubated with corresponding Alexa Fluor conjugated secondary antibodies (Invitrogen) in a 1:1000 dilution in 1X PBS for 1h30 at room temperature. After washing, sections were counterstained with DAPI and mounted on glass slides with ProLong Gold Antifade Mountant (ThermoFisher).

Puromycin-proximity ligation assay (Puro-PLA)

Puro-PLA experiments were performed as previously described[62] to label newly synthesized β-actin, TUBB2B, and APC proteins. Briefly, neurons cultured 6 days *in vitro* were incubated with 10μg/mL puromycin for 5min, washed with warm PBS and fixed for 10 min in 4% PFA-sucrose. Following fixation and permeabilization, the Duolink assay (DUO92013) was performed as recommended by the manufacturer using the following primary antibodies: Puromycin (Merck Millipore, MABE343, 1:400), APC (Santa Cruz, sc-896, 1:200), β-actin (Cayman Chemicals, 32127, 1:200) and TUBB2B (Abcepta, AP11940a). Cells were mounted in ProLong Diamond Antifade Mountant (ThermoFisher) and images were acquired with a Dragonfly confocal microscope (Oxford Instruments).

RNA fluorescence *in situ* hybridization (FISH) on cultured mouse neurons

DIV6 cultured neurons were fixed and processed using the ViewRNA Cell Plus Assay kit (ThermoFisher) based on the manufacturer’s instructions. Custom-made ViewRNA Cell Plus Probe Sets (ThermoFisher) against mouse *Actb* mRNA (VB6-12823-VC) and mouse *Apc* mRNA (VB6-19724-VC) were used. YTHDF1 staining was performed during the ViewRNA Cell Plus Assay experiment according to the manufacturer’s protocol. Neurons were then counterstained with DAPI and observed under a Dragonfly confocal microscope (Oxford Instruments).

EB3 comets analysis *in vitro*

DIV2 neurons cultured in 35 mm glass-bottom dishes (MatTek) were transfected with EB3-GFP plasmid and either shControl and sh*Ythdf1* plasmids and imaging was performed at DIV6. During imaging, the dishes were maintained in the incubation chamber of the LSM800 laser scanning confocal microscope at 37°C with a continuous supply of 5% CO_2. Images were acquired with 63× Plan Apochromat objective using ZEN software (Zeiss). Axonal growth cone images were captured every 3 s during 100 s. Kymographs were generated using the ImageJ plugin KymoToolBox (fabrice.cordelieres@curie.u-psud.fr).

Live-imaging of β-actin and Tubb2b mRNA

DIV2 neurons cultured in 35 mm glass-bottom dishes (MatTek) were transfected with the reporter construct (5′UTR-SunTag24x-Actin-3′UTR-PP724x or 5′UTR-SunTag24x-Tubb2b-3′UTR-PP724x), PP7-mCherry plasmids and either shControl, sh*Ythdf1 or* sh*Apc* plasmids and imaging was performed at DIV6. During imaging, the dishes were maintained in the incubation chamber of the LSM800 laser scanning confocal microscope at 37°C with a continuous supply of 5% CO_2. Axon images were captured at 500 ms intervals for 120 s under a Dragonfly confocal microscope (100× magnification, Oxford Instruments). Kymographs were generated using the ImageJ plugin KymoToolBox (fabrice.cordelieres@curie.u-psud.fr). β-actin mRNA granules were considered as stationary if the directional transport observed was less than 4μm within 120 s. Live imaging was conducted approximately 50 μm away from the cell soma, with the analysis covering a length of about 100 μm. Average speed refers to the total distance traveled by a β-actin granule divided by the overall time, including periods of pausing. In contrast, instantaneous speed is calculated as the total distance traveled divided by the time spent moving, excluding any pausing time. Run length is the total distance traveled by every moving β-actin granules over 120 s. Pausing time is defined as the percentage of total travel time during which the organelle's velocity drops below 0.1 μm/s β-actin mRNA granules were considered as moving if the directional transport observed was more than 4μm within 120 s.

Western blotting

Neuro2a cell lysates were prepared using a RIPA buffer after washing with ice-cold PBS. RIPA buffer composition was as follows: 50 Mm Tris-Cl pH 7.6, 150 mM NaCl, 1 mM EDTA (pH 8.0), 1% Triton X-100, 4% SDS and Ultrapure water. For 10 mL of RIPA buffer, 1 tablet of the protease inhibitor cocktail from Roche was added. The cell suspension was sonicated using TOMY Ultrasonic Disruptor UD-201. The suspension was centrifuged at 15000 RPM for 5 min at 4°C using a HIMAC CT15RE benchtop centrifuge. The supernatant was collected in a low-binding Eppendorf tube and further subjected to BCA assay using Pierce BCA Protein Assay Kit (ThermoFisher) according to the manufacturer's protocol. Subsequently, lysate samples were boiled with 5X SDS at 95°C and 20 μg of lysates were loaded on 10% SDS–polyacrylamide gel to be finally transferred to a PVDF membrane for immunoblotting. To develop HRP chemiluminescence signal Amersham ECL Select Western Blotting Detection Reagent (Cytiva, Merck) was used in 1:1 dilution. Membranes were imaged using BioRAD GelDoc XR Molecular Imager. Band densitometry was performed using ImageJ.

qPCR analysis

Transfected Neuro-2a cells were processed for RNA extraction using RNeasy Mini Kit (Qiagen) following the manufacturer's instructions. 1μg of RNA was reverse-transcribed with iScript RT Supermix (Bio-Rad). Following this, appropriate dilution of the cDNA was then used in qPCR reactions containing PowerUp SYBR Green Master Mix (Applied Biosystems, Thermofisher) and primer pairs. qPCR reactions were performed using the StepOnePlus Real-Time PCR System (Applied Biosystems). Analysis was done using the 2-ΔΔCT method after defining primer efficiencies, *Gapdh* served for normalization. The following primers were used: *Ythdf1*, GCATCAGAAGGATGCAGTTCATG (forward) and GATGGTGGATAGTAACTGGACAG (reverse); *Mettl14,* AGAGTGCGGATAGCATT GGTGC (forward) and CTCCTTCATCCAGACACTTCCG (reverse); *Apc,* CCCCAGTGACCTTCCAGATA (forward) and GCACTGT CTGTGGAGGAGGT (reverse).

RIP-qPCR

Neuro2a cells were plated 6-well plates and were transfected with 2.5 μg of DNA per well using Lipofectamine 2000. Transfected and differentiated Neuro2a cells were lysed in RIP buffer (25mM Tris–HCl pH 7.5, 150 mM NaCl, 2.5 mM EDTA, 0.5 mM DTT, 0.5% Triton X-100, RNase inhibitor, EDTA-free protease inhibitor). The harvested lysates were mixed for 20 min at 4°C. Concomitantly, 2 μg of anti-YTHDF1 antibody was added to 30μL Dynabeads Protein G per sample in PBS containing 0.005% Tween and incubated at 4°C for 1hr. Lysates were cleared by centrifugation at 15,000 rpm for 10 min. One-third of supernatant was kept for total protein and RNA analysis. The remaining of the supernatant was transferred to the tube with the antibody-coupled beads for 2 h at 4°C. Following immunoprecipitation, beads were washed three times with a lysis buffer containing 0.01% Triton X-100. One-fifth of the beads were used for protein elution in a 5× sample buffer for western blot analysis. The rest of the beads were resuspended in 300mL of Trizol, and RNA was extracted with Direct-zol RNA MiniPrep Kit. cDNA was prepared with Superscript III and RT-qPCR was performed using NZYSupreme qPCR Green Master Mix (2x) in QuantStudio5 qPCR machine. Enriched *Apc* mRNA binding to YTHDF1 was normalized to input RNA.

SELECT assay for single-base m6A detection

m^6A coordinates on *Apc* mRNA were selected from various single nucleotide m^6A sequencing studies like DART-seq,[83] m^6A CLIP-seq using mouse brain samples,[45,59] YTHDF1 binding site analysis[45] and our own m^6A iCLIP dataset from mouse forebrain. All of these sites were also corroborated in m^6A Atlas.[84] To assess m^6A modification levels at the selected individual sites, we designed four probe pairs specifically targeting the A7836, A11110, A11121 and A12324 sites within the *Apc* transcript (GenBank: NM_007462.3) for SELECT assays. The SELECT method was performed as described previously.[60] In brief, 2.2μg of total RNA from Neuro2a cells was combined with 40 nM Up Primer, 40 nM Down Primer, and 5 μM dNTP in a 17μL solution of 1×CutSmart buffer (containing 50 mM KAc, 20 mM Tris-HAc, 10 mM MgAc2, 100 μg/mL BSA, pH 7.9 at 25°C). The RNA and primers were then annealed by subjecting the mixture to a temperature gradient: 90°C for 1min, 80°C for 1min, 70°C for 1min, 60°C for 1min, 50°C for 1min, and finally 40°C for 6min. Following annealing, a 3 μL mixture comprising 0.01 U Bst 2.0 DNA polymerase, 0.5 U SplintR ligase, and 10 nmol ATP was added to the previous mixture, bringing the final volume to 20 μL. The reaction mixture was then incubated at 40°C for 20 min, denatured at 80°C for 20 min, and finally kept at 4°C. Following this, quantitative qPCR reactions were conducted using the StepOnePlus Real-Time PCR System (Applied Biosystems, Thermofisher). The 20μL qPCR reaction mixture comprised 2x PowerUp SYBR Green Master Mix (Applied Biosystems, Thermofisher), 200 nM of qPCR-F primer, 200 nM of qPCR-R primer, 2μL of the final reaction mixture, and ddH2O. The qPCR protocol was performed using the following steps: 95°C for 5 min; (95°C for 10 s, 60°C for 35 s) repeated for 40 cycles; 95°C for 15 s, 60°C for 1 min; 95°C for 15 s; and finally, holding at 4°C. Data analysis was performed using StepOne Real-Time PCR Software v2.3.

Analysis of neuronal morphology

Fluorescence images of DIV6 neurons to study axonal growth were acquired on a BZ-X700 microscope (Nikon). Multichannel images were acquired to cover the entire territory of single neurons and stitched in the BZ-X viewer to capture the entire axon length. The Segmented Line selection tool in ImageJ software was used for tracing axons and quantifying the axon length. Two parameters were measured to assess axonal growth: 1. length of the longest axon. 2. The number of parallel branches (by counting the number of the endpoints of the axonal branches, excluding thin and short protrusions). Representative neurons were traced using the Simple Neurite Tracer (SNT) plug-in (ImageJ), and processed in Paint3D and vectorized for visualization.

QUANTIFICATION AND STATISTICAL ANALYSIS

To perform all the statistical analysis GraphPad Prism version 8 was used. The n number of each experiment and a detailed description of the method of statistical analysis and the results are mentioned in the legend of each figure. Comparison between two groups with normally distributed samples was performed with a two-sided unpaired t test, while comparisons among more than two groups with one-way analysis of variance (ANOVA) followed by the appropriate post hoc test. The statistical notations in the graphs are as follows: ns, not significant; *$p \leq 0.05$; **$p \leq 0.01$; ***$p \leq 0.001$; ****$p \leq 0.0001$. p-value >0.05 is considered non-significant.